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Atamer Erkal

Healthcare & Defence AI Engineer
MLOps & Explainable AI Specialist

Former NATO Planning Officer | Southern Germany (Available)
EU Work Permit

PROFIL

Former NATO Planning Officer and Healthcare & Defence AI Engineer with over 15 years of experience in mission-critical and highly regulated environments. Specialized in the development of production-grade end-to-end ML pipelines using MLflow, DVC, Docker, and Azure ML, with deep expertise in Explainable AI (SHAP, LIME, Grad-CAM), model monitoring, and full compliance with GDPR and EU AI Act.

Proven experience in Healthcare AI (medical imaging, voice-based triage, clinical NLP analysis) and Defence AI (air track classification, real-time AI edge systems). Trained over 100 data scientists with a 90% placement rate.

TECHNICAL SKILLS

- **Programming & Data**
  Python (pandas, NumPy, scikit-learn), SQL (PostgreSQL), Git
- **Machine Learning & AI**
  Supervised & unsupervised learning (classification, regression, clustering), Deep Learning (CNNs, EfficientNet, YOLOv8 with PyTorch & TensorFlow), Computer Vision & Medical Imaging, Generative AI & RAG, Model evaluation, validation and optimization, Explainable AI (SHAP, LIME, Grad-CAM), Voice AI & Speech Processing)
- **MLOps & Cloud**
  MLflow, DVC, GitHub Actions, Docker (containerization & CI/CD), Azure ML (cloud deployment, pipeline automation, monitoring), ETL pipelines, REST APIs (FastAPI), TensorRT (Edge Inference)
- **Data Visualization & BI**
  Power BI, Matplotlib, Seaborn, Plotly – interactive dashboards for analyses
- **Domain & Regulation**
  Healthcare (FHIR/HL7, clinical decision support, PII redaction), Defence (NATO-compatible systems, real-time edge AI), GDPR for Data Science & AI, EU AI Act (risk classification & compliance)

PROFESSIONAL EXPERIENCE

Program Director & Machine Learning Engineer
Clarusway – USA & Germany | *04/2020 – present*

- *Design and implementation of production-oriented ML and MLOps pipelines using MLflow, DVC, and Azure ML*
- *Leadership of end-to-end Machine Learning projects with focus on monitoring, retraining, model stability, and explainability in regulated environments*
- *Development of explainable AI solutions for imaging and real-time edge systems*
- *Mentoring and training of 100 participants in Python, Machine Learning; placement rate approx. 90% in Data roles*
- *Close collaboration with international stakeholders to translate business requirements into viable, GDPR- and EU AI Act-compliant ML solutions*
- ***Award:** Best Data Science & Analytics Program (2021–2022).*

Transition Phase and Full-time Upskilling in Data Science (Further Training etc.) | *2017 -2020*

Planning Officer & Electrical Engineer – Safety-Critical Systems
NATO and Naval Headquarters – United Kingdom & Turkey | *2002 – 2017*

- *Worked in multinational, highly regulated, and security-critical environments with data-based decision support*
- *Conducting analytical evaluations for resource allocation and performance assessment*
- *Application of structured quantitative methods under time and security constraints*
- ***Award: C**ertificate of Honour for Leadership and Management (Naval Forces Commander)*

SELECTED PROJECTS

(Further projects and live demos available on my portfolio: *atamererkal.github.io* – Scan the QR code or visit directly)

- ***VitalNav.ai – Voice-First AI Triage System (Healthcare - Flagship)***
  Development of an innovative voice-based pre-hospital emergency triage system using Azure OpenAI (GPT-5.4 mini), RAG, Azure AI Speech Services, Azure AI Translator, and Azure Maps. Triage time under 5 seconds, support for 3 languages, fully GDPR compliant.

- ***SkinxAI – Explainable Medical Image Classification (Healthcare)***
  CNN model (EfficientNet-B7) for skin lesion detection achieving 85.2% sensitivity for malignant cases. Grad-CAM visualization and risk stratification (HIGH/MEDIUM/LOW).

- ***AEGIS-Cloud – Autonomous AI Edge Guardian (Defence - Flagship)***
  Development of a real-time edge AI system for ship detection and classification using YOLOv8 and TensorRT (FP16 optimization). Achieved latency reduction to under 30 ms on resource-constrained edge devices. Integration of Moondream VLM for visual reasoning and full EU AI Act-compliant audit trails for transparency in safety-critical defence applications.

- ***VANGUARD – Air Track Classification for NATO C2 (Defence)***
  XGBoost-based classification system for air tracks (Hostile/Friendly/Civilian) with 96.3% accuracy, 3D visualization, and NATO-compatible architecture.

- ***Explainable AI Dashboard (Responsible AI)***
  Interactive dashboard for model interpretation using SHAP and LIME to support transparency and traceability for high-risk AI systems according to the EU AI Act.

VOLUNTEER WORK

German Red Cross (DRK) & Support Facilities – Ulm Area

- *Support in multilingual communication and daily organization.*

EDUCATION

- **M.A. Management & Leadership**
  National Defence University, Istanbul | 2010–2012
  Award: Winner of the university-wide article competition (Thesis / Management Essay)

- **B.Sc. Electrical Engineering & Officer Training**
  National Defence University, Istanbul | 1998–2002

CERTIFICATES & FURTHER TRAINING

Further Training:

- *DataJarLabs: Data Science and Analytics | 2018–2019*
- *NATO School (Germany): Comprehensive Operations Planning*
- *Ankara University: International Relations (Further Qualification)*

Certificates:

- *Microsoft AI-102 (AI Engineer Associate Certificate) (planned May 2026)*
- *Microsoft: AI on Azure, Azure ML, Power BI & Excel for Data Analysis*
- *DataCamp: Data Scientist with Python, MLOps, CI/CD for ML*
- *Agile with Jira (Atlassian)*
- *Power BI & Excel für Datenanalyse*

Regulatory Further Training (ongoing):

- *GDPR for Data Science & AI*
- *EU AI Act – Risk Classification & Compliance*

LANGUAGES

Turkish (Native), English (Fluent), German (B2 – TELC, Exam 04/2025)

REFERENCES

Available upon request (including Clarusway, NATO contacts)